OutburstMapper demo

This guide walks through loading a saved ROI session, opening outburst and surface images, and inspecting existing boundaries in spacecraft view with linked zoom. We use the August 1, 2015 outburst associated with ROI 8 as an example. A button guide explains the controls used throughout the app.

1. Start the app and load the saved session

Install the software from GitHub and unpack the Zenodo data into the repository's **data** folder. From the OutburstMapper folder, run:

```
python outburst_mapper.py
```

The shape model and SPICE kernels load automatically. Check for **SPICE: loaded (ROS_OPS.TM)**. Scroll down the left sidebar, click **Load session...**, and open:

```
data/sessions/67P_outburst_session.json
```

The session restores all 78 ROI entries. Keep them enabled initially, or click **Select all**, to inspect the complete set. Rotate the model to see outlines on other sides.

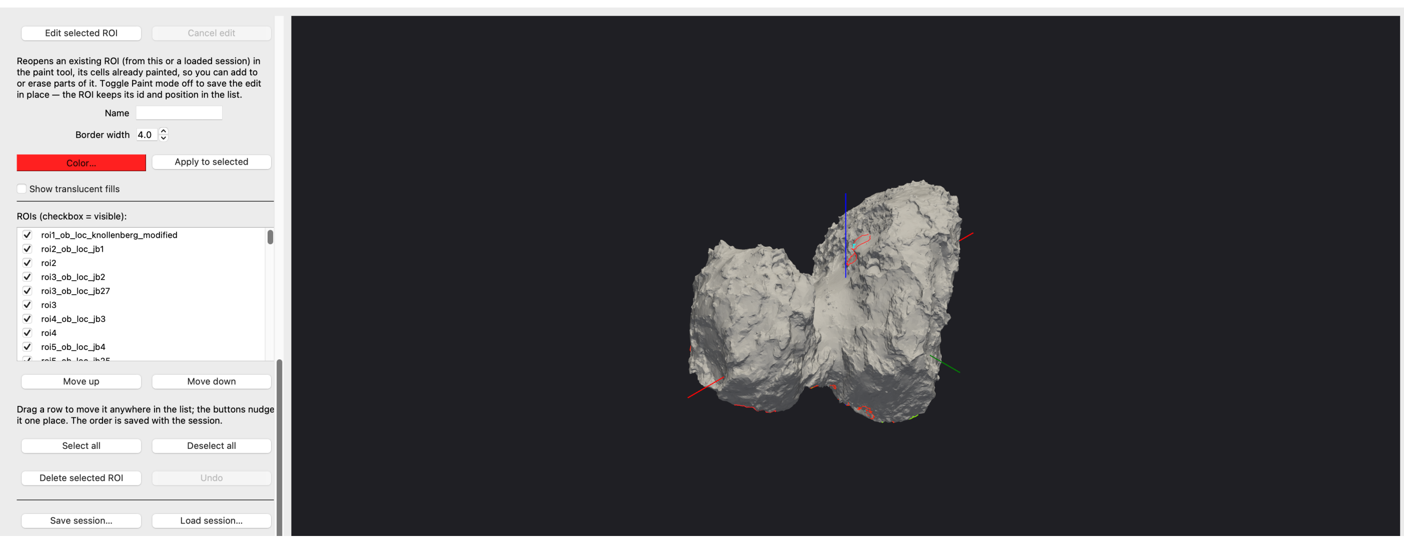


Figure 1. All saved ROI entries enabled after loading the session. The list and session controls are at left; the visible outlines depend on the model viewpoint.

2. Keep only the ROI 8 boundaries

Click **Deselect all**, then tick these two entries:

`roi8_ob_loc_jb7`	August 1, 2015 outburst source (red)
`roi8`	Surface-change region (cyan)

ROI 8 is a broader region consisting of several features. Here we focus on the sub-region associated with the August 1, 2015 outburst.

3. Load the outburst image and match the view

Click **Load image (.cub/.IMG/.LBL)...** and choose the file below. Click **View from spacecraft**. Keep **Link zoom** checked, then zoom and pan toward the ROI 8 outlines to reduce empty background.

```
data/ROI_data/roi8/outburst_img/
N20150801T105157775ID4EF22.cub
```

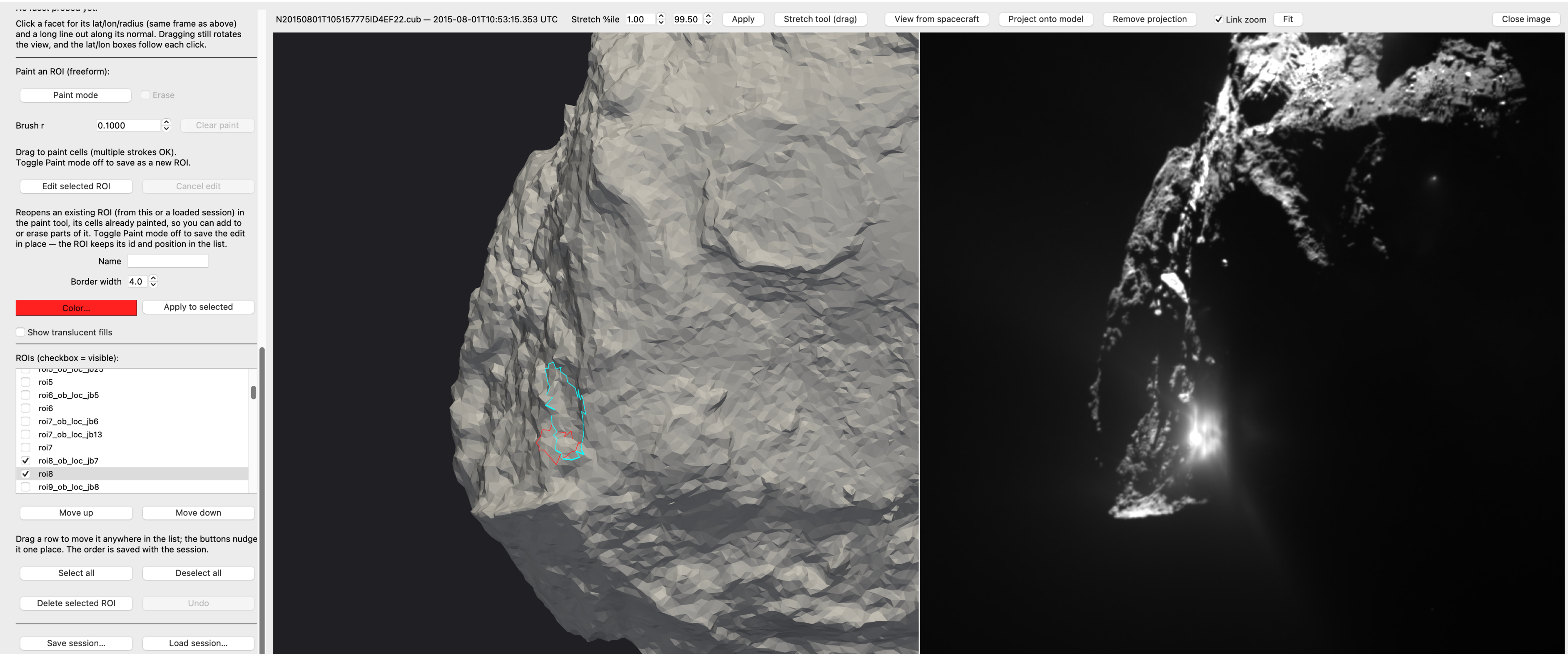


Figure 2. August 1, 2015, 10:53:15.353 UTC. The model (left) and outburst image (right) use the same spacecraft geometry and linked framing. The red outline marks the saved August 1 outburst-source footprint; cyan marks the broader surface-change region. The August 1 plume is the event illustrated here.

Viewing tip. Use the wheel to zoom and drag to pan while linked. Adjust **Stretch %ile** and **Apply** for contrast; **Fit** returns to the full image. A footprint is a saved ROI boundary: **Project onto model** is optional and is not used in these views.

4. Load the surface image and compare

Use **Load image...** to open the surface image below, then click **View from spacecraft again** to use this image's own time and instrument. Keep the same two ROI 8 entries enabled and use linked zoom to inspect the terrain.

```
data/ROI_data/roi8/surface_img/
N20160123T224510679ID4EF24.cub
```

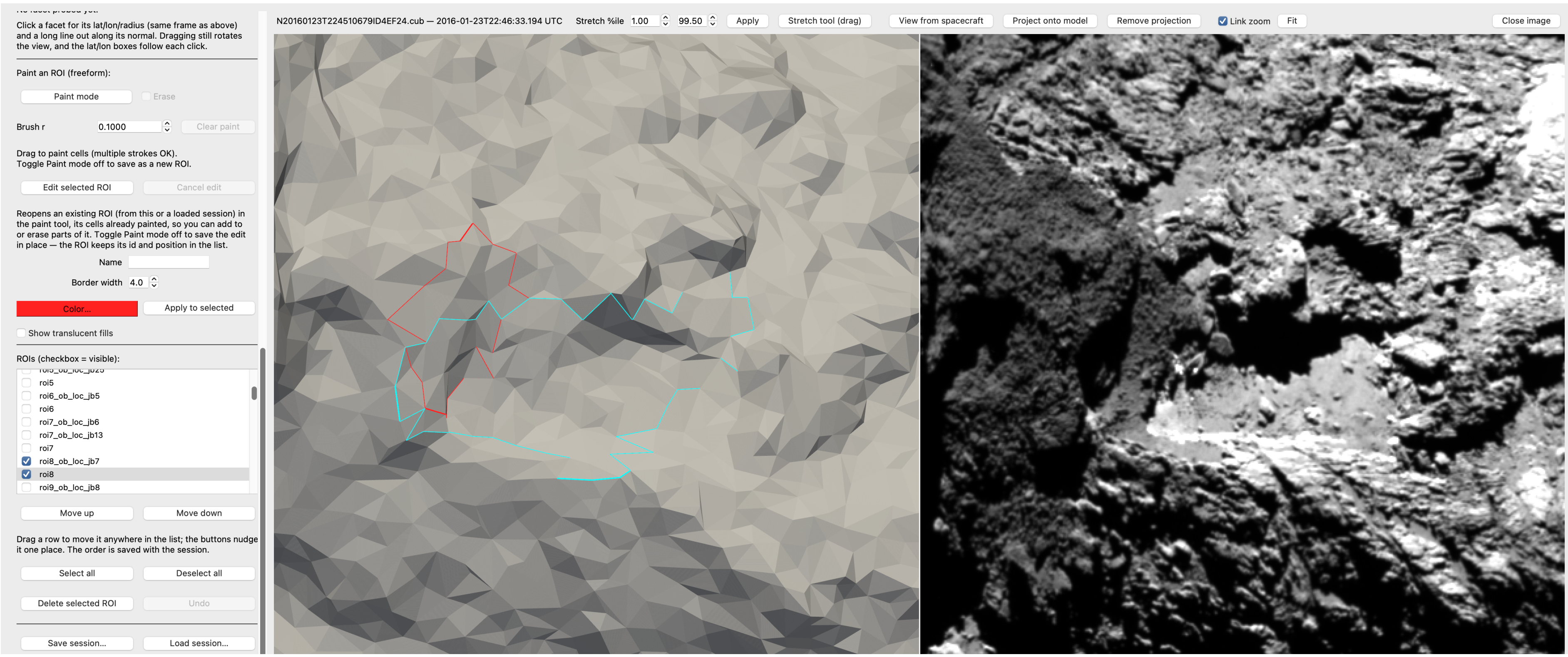
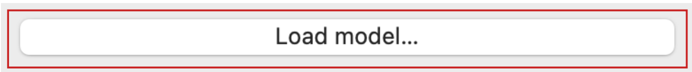


Figure 3. January 23, 2016, 22:46:33.194 UTC. The cyan outline shows the saved surface-change region, with the red August 1 source footprint retained for comparison. The model is shaded for viewing, so its brightness and shadows do not reproduce the photograph. The two images in this demo are different observations, not a before/after image pair.

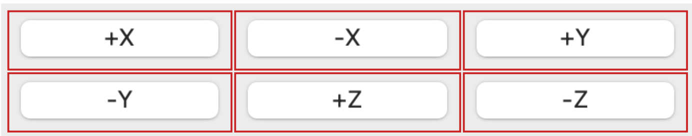
Save or revisit. Save screenshot... exports the 3D view only. Capture the whole application to retain both panels. **Save session...** stores ROI definitions and visibility; reopen the image and choose **View from spacecraft** again in a later session.

Button guide: model and spacecraft view

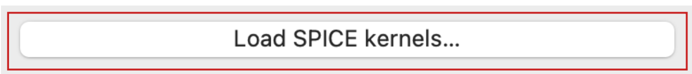
The sidebar scrolls vertically. Red outlines identify the actual controls; arrows connect them to their descriptions.



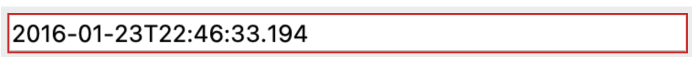
Load model...
Open an OBJ, PLY, STL, VTK, or VTP shape model. Loading another model clears the current ROI list and overlays.



+X / -X / +Y / -Y / +Z / -Z
Snap to a view along a body-fixed model axis. Use View from spacecraft to return to the image's observing geometry.

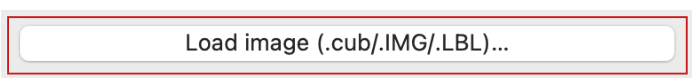


Load SPICE kernels...
Choose a metakernel describing the geometry kernels. The status below reports whether loading succeeded.



Date/time (UTC)
Enter an observing time for the manual instrument-view buttons.

NAC / WAC / NAVCAM
Set the model camera to the selected instrument at the entered UTC time. For a loaded image, View from spacecraft reads these details automatically.



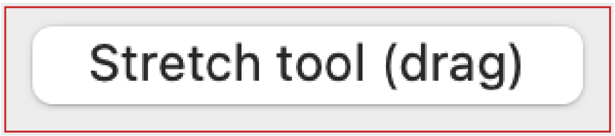
Load image (.cub/.IMG/.LBL)...
Open a mission image beside the model. For detached-label PDS3 products, select the LBL file.

Show projected image
Show or hide an image that has already been projected onto the model. This does not control ROI footprints.

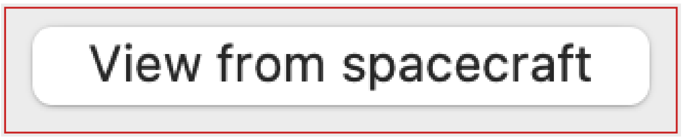
Button guide: image toolbar

This toolbar appears when an image is loaded. It spans the model and image panels.

Stretch %ile / Apply
Set the lower and upper percentiles used for image contrast, then apply them. The image data remain unchanged.



Stretch tool (drag)
Enable, then draw a rectangle in the image to set contrast from that region's minimum and maximum valid values. Disable it to resume normal panning.

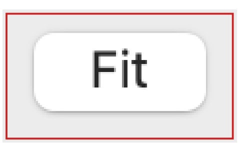


View from spacecraft
Read the loaded image's instrument and acquisition time, place the model camera at that viewpoint, and link the views.

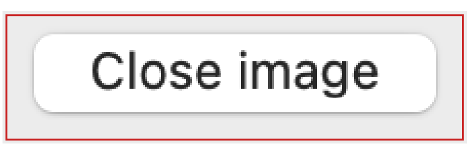
Project onto model
Drape the loaded image onto the terrain using SPICE camera geometry and visibility testing.

Remove projection
Remove the image texture from the model; the loaded image panel remains available.

Link zoom
Synchronize pan and zoom between the image and model. Uncheck to navigate the 3D model independently.



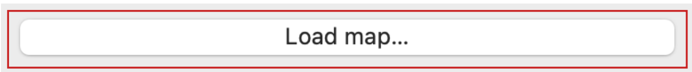
Fit
Fit the whole image into its panel; with linked views, the model framing follows.



Close image
Hide the image panel and restore a single 3D view. An existing projection stays until Remove projection is used.

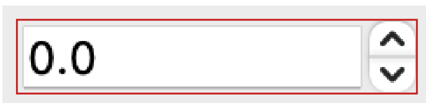
Button guide: maps, display, and locations

These optional controls help orient the model and inspect locations. They are not required for the ROI 8 walkthrough.

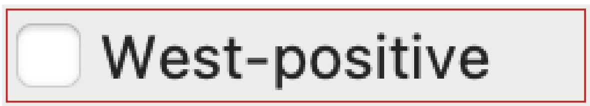


Load map...
Drape an equirectangular longitude/latitude map over the model.

Show map / Remove map
Toggle the loaded map's visibility, or remove it entirely.

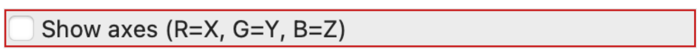


Lon at centre
Set the longitude at the map's central column. The two supplied maps use 0 degrees.

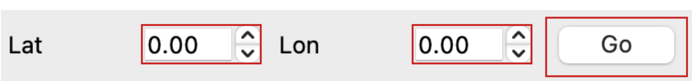


West-positive
Enable for maps whose longitude increases westward. Leave unchecked for the supplied maps.

Show plate edges
Show the edges of the shape model's triangular facets.

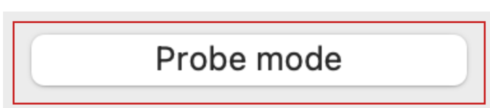


Show axes
Show model X, Y, and Z axes in red, green, and blue.

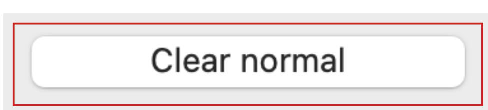


Lat / Lon / Go
Enter planetocentric latitude and east-positive longitude, then centre the view there and add a location marker.

Show marker
Show or hide the location marker.



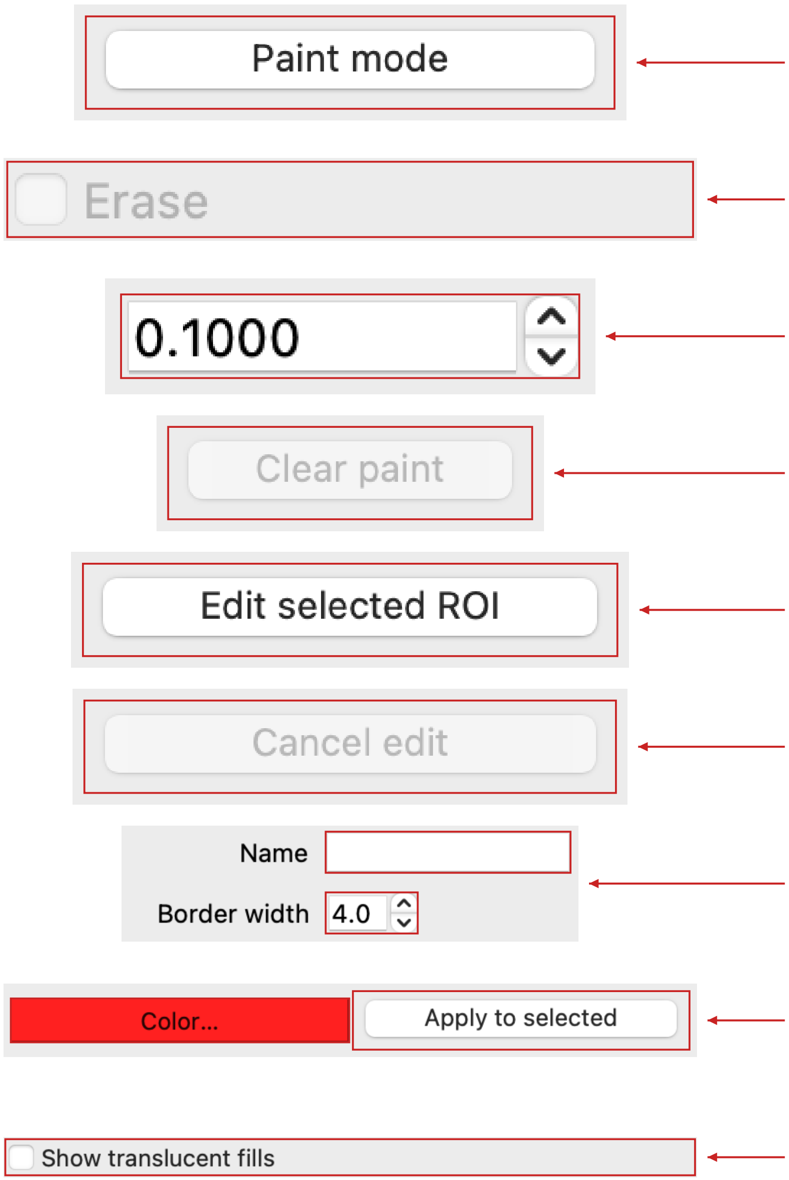
Probe mode
Click a facet to report latitude, longitude, radius, and outward normal. Dragging still navigates.



Clear normal
Remove the probe's normal line and readout.

Button guide: painting and ROI appearance

Reference only: this demo does not draw or edit footprints. Work on a copy of the supplied session before making edits.



Paint mode
Drag to paint facets. Toggle off to finish a new ROI, or to save an edit of the selected ROI.

Erase
Remove painted facets while paint mode is active.

Brush r
Set the brush radius in model units (kilometres for the supplied shape model).

Clear paint
Clear the current in-progress painted selection.

Edit selected ROI
Reopen the selected footprint for painting or erasing. Finishing updates that same ROI.

Cancel edit
Discard the in-progress edit and retain the saved ROI.

Name / Border width
Set a new ROI's name and border width, or prepare changes for the selected ROI.

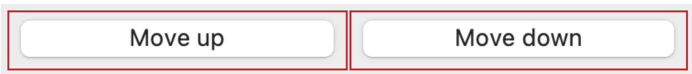
Color... / Apply to selected
Choose the colour. Apply to selected writes the current name, colour, and width to the selected ROI and auto-saves the session.

Show translucent fills
Add translucent colour inside the visible footprint outlines.

Button guide: ROI list and saving

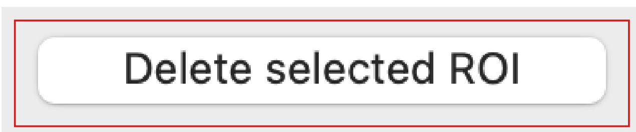
ROI edits auto-save to the loaded session. Image files, contrast settings, map textures, and camera views are not stored in the session JSON.

ROI list checkboxes
Show or hide individual footprints. Clicking a row selects it for editing; its checkbox controls visibility.

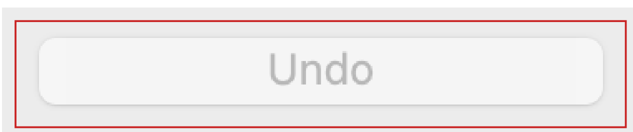


Move up / Move down
Move the selected ROI one place in the list. Dragging a row can also reorder it; the order is saved.

Select all / Deselect all
Show or hide every footprint at once.

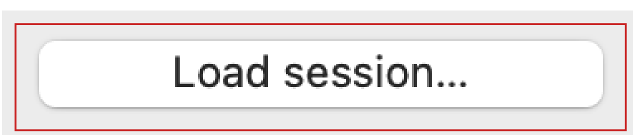


Delete selected ROI
Delete the selected footprint and auto-save the session.

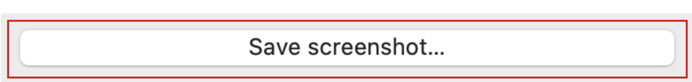


Undo
Restore the previous ROI state after creation, a finished edit, an applied style change, or deletion. Up to 20 changes are retained.

Save session...
Save the model reference, ROI facets, names, colours, border widths, order, and visibility to JSON. The selected file becomes the target for later auto-saves.



Load session...
Restore a saved model and ROI list. The session does not restore the loaded image, its contrast, or its spacecraft view.



Save screenshot...
Export the current 3D view to PNG. It does not include the side-by-side image panel or the controls.